Homework 1

2.1.1.a. Let μ be a positive inner regular locally finite measure. Suppose there exists a constant D such that for all $x \in \mathbb{R}^n$ and $r > 0$ we have

$$\mu(3B(x,r)) \leq D\mu(B(x,r)).$$

Define the weak $L^{1,\infty}$ norm $\|\cdot\|_{L^{1,\infty}}$ by

$$\|f\|_{L^{1,\infty}} = \sup_{\lambda>0} \lambda\mu(|f| > \lambda).$$

We will show the uncentered maximal function M is continuous $L^1(\mathbb{R}^n, \mu) \to L^{1,\infty}(\mathbb{R}^n, \mu)$. Define $E_\alpha = \{x \in \mathbb{R}^n : Mf(x) > \alpha\}$. We need to show

$$\|Mf\|_{L^{1,\infty}} \leq C\|f\|_{L^1},$$

and it will suffice to show for some $\alpha > 0$ that

$$\alpha\mu(E_\alpha) \leq C\|f\|_{L^1}.$$

Now use inner regularity. Choose any compact $K \subset E_\alpha$ and cover K in balls $\{B_x\}_{x\in K}$ such the average value of f on every B_x is greater than α. Choose a finite subcover $\{B_i\}_1^N$ and choose a further subcover $\{B_{i_k}\}_1^M$ such that $\cup B_i \subset \cup 3B_{i_k}$. Then

$$\mu(K) \leq \mu\left(\bigcup_{k=1}^{M} 3B_{i_k}\right) \leq D\sum \mu(B_{i_k}) \leq \frac{D}{\alpha}\sum \int_{B_{i_k}} |f(y)|dy \leq \frac{D}{\alpha}\|f\|_{L^1}.$$

2.1.2.a. Let $\mathscr{F}$ be a finite family of open intervals in $\mathbb{R}$. We prove there exist two subfamilies $\mathscr{F}_1$ and $\mathscr{F}_2$ of $\mathscr{F}$, each family pairwise disjoint, such that the union of the sets in both families is equal to the union of $\mathscr{F}$.

Without loss of generality, we can reduce to the case when $\cup\mathscr{F}$ is connected, and we can remove any intervals strictly contained in larger ones. If the union is not disjoint, we can just solve the problem on disjoint pieces and combine the individual families in any way.

We proceed by induction. The case of $n = 1$ is trivial. Now assume we have proven that for any family of $k-1$ elements, we can find two disjoint subfamilies as above.

Let $\mathscr{F}$ be a connected family of k intervals. Remove the leftmost set U_0, which must exist since $\mathscr{F}$ is finite, and we have removed any sets contained in others. Then from $\mathscr{F} \setminus \{U_0\}$, choose subfamilies $\mathscr{F}_i$, $i = 1, 2$ which satisfy the desired property.

Remove elements of both subfamilies entirely contained in U_0. Then I claim we have two cases:

1. U_0 intersects only one of the subfamilies
2. U_0 intersects both subfamilies.

This is because U_0 cannot intersect multiple elements of the same subfamily: U_0 is an interval, so the pairwise disjointness of the subfamilies would force the leftmost of the two sets to be contained entirely in U_0. Then the only cases are U_0 meets one family, both, or none. But it cannot meet none as this would imply $\mathscr{F}$ was disjoint.

In case (1.), we can just add U_0 to the subfamily it doesnt intersect.

For case (2.), suppose U_0 meets the sets $V \in \mathscr{F}_1$ and $W \in \mathscr{F}_2$. We know $V \cap W \neq \varnothing$, and suppose without loss of generality that V is to the left of W. Then since U_0 intersects W, and U_0 extends left past V, V is contained in the union of U_0 and W. Thus, we can replace V in $\mathscr{F}_1$ by U_0. This completes the proof.

Using this result, we prove the uncentered maximal function M of the previous exercise map $L^1(\mu) \to L^{1,\infty}(\mu)$ with constant 2.

Define $E_\alpha = \{x \in \mathbb{R}^n : Mf(x) > \alpha\}$. Again cover E_α in balls on which the average value of f is greater than α. Choose any compact $K \subset E_\alpha$ and choose a finite subcover $\mathscr{F}$ of K. Choose subfamilies $\mathscr{F}_1$ and $\mathscr{F}_2$ which have the property above. Now

$$\mu(K) \leq \mu\left(\bigcup_{B \in \mathscr{F}} B\right) \leq \mu\left(\bigcup_{B \in \mathscr{F}_1} B\right) + \mu\left(\bigcup_{B \in \mathscr{F}_2} B\right) \tag{1}$$

$$= \sum_{B \in \mathscr{F}_1} \mu(B) + \sum_{B \in \mathscr{F}_2} \mu(B) \leq \frac{1}{\alpha}\left(\int_{\mathbb{R}} |f| + \int_{\mathbb{R}} |f|\right) = \frac{2}{\alpha}\|f\|_{L^1}, \tag{2}$$

where the last inequality follows exactly as in (2.1.1). Taking a supremum over all compact K, we obtain the result.

2.1.3. Define $\mathscr{M}_c$ and M_c, the centered and uncentered Hardy-Littlewood maximal operators, with the supremum taken over all cubes with sides parallel to the axes in $\mathbb{R}^n$ instead of balls. We prove

$$(a.) \quad 1 \leq \frac{M(f)}{\mathscr{M}(f)} \leq 2^n, \quad (b.) \quad \frac{1}{n^{\frac{n}{2}}}\frac{2^n}{v_n} \leq \frac{M(f)}{M_c(f)} \leq \frac{2^n}{v_n}, (c.) \quad \frac{1}{n^{\frac{n}{2}}}\frac{2^n}{v_n} \leq \frac{\mathscr{M}(f)}{\mathscr{M}_c(f)} \leq \frac{2^n}{v_n}$$

where v_n is the volume of the n-dimensional unit ball.

a. We have

$$M(f) = \sup_{r>0}\sup_{B_r \ni x} \frac{1}{\mu(B_r)}\int_{B_r} |f(y)|d\mu \leq \sup_{r>0} \frac{1}{\mu(B_r)}\int_{B_{2r}(x)} |f(y)|d\mu,$$

since $|f|$ is positive. Moreover, we have $\mu(B_{2r}) = 2^n\mu(B_r)$. Thus

$$\cdots \leq \sup_{r>0} 2^n \frac{1}{\mu(B_{2r})}\int_{B_{2r}(x)} |f|d\mu = 2^n \sup_{r>0}\frac{1}{\mu(B_r)}\int_{B_r(x)} |f|d\mu = 2^n\mathscr{M}(f).$$

We already know that $\mathscr{M}(f) \leq M(f)$, since the supremum in the uncentered operator is taken over a larger set.

b. Let R_r be a cube of side length $2r$, and note that the smallest concentric circle containing B_r has radius $\sqrt{nr^2} = r\sqrt{n}$. Now

$$M_c(f) = \sup_{r>0}\sup_{R_r \ni x}\frac{1}{\mu(R_r)}\int_{R_r} |f|d\mu \leq \sup_{r>0}\sup_{R_r \ni x}\frac{1}{\mu(R_r)}\int_{B_{r\sqrt{n}}} |f|d\mu.$$

We have $\mu(R_r) = 2^n r^n$, and $\mu(B_{r\sqrt{n}}) = n^{\frac{n}{2}} r^n v_n$. Thus

$$\frac{1}{\mu(R_r)} = \frac{n^{\frac{n}{2}}v_n}{2^n}\frac{1}{\mu(B_r)}.$$

Finally, after noting that the set $\{R_r : x \in R_r\}$ can be mapped injectively to $\{B_{r\sqrt{n}} : x \in B_{r\sqrt{n}}$, we have

$$\cdots \leq \sup_{r>0}\sup_{R_r \ni x}\frac{n^{\frac{n}{2}}v_n}{2^n}\frac{1}{\mu(B_r)}\int_{B_{r\sqrt{n}}} |f|d\mu \leq \sup_{r>0}\sup_{B_r \ni x}\frac{n^{\frac{n}{2}}v_n}{2^n}\frac{1}{\mu(B_r)}\int_{B_r} |f|d\mu = \frac{n^{\frac{n}{2}}v_n}{2^n}M(f).$$

For the next inequality, note that the smallest cube containing B_r is just R_r. We have

$$\frac{1}{\mu(B_r)} = \frac{2^n}{v_n}\frac{1}{\mu(R_r)},$$

so that

$$M(f) = \sup_{r>0} \sup_{B_r \ni x} \frac{1}{\mu(B_r)} \int_{B_r} |f| d\mu \leq \sup_r \sup_{B_r} \frac{1}{\mu(B_r)} \int_{R_r} |f| d\mu = \sup_r \sup_{B_r} \frac{2^n}{v_n} \frac{1}{\mu(R_r)} \int_{R_r} |f| d\mu = \frac{2^n}{v_n} M_c(f).$$

c. The last case is exactly the same as above, except there is no second supremum taken over all balls containing x.

2.1.8. Let f_0 be the characteristic function on the unit ball in $\mathbb{R}^n$. For $|x| > 1$, define

$$B_x = B(\frac{1}{2}(|x| - |x|^{-1}) \frac{x}{|x|}, \frac{1}{2}(|x| + |x|^{-1})).$$

Any such ball always intersects more than half of the unit ball: We can assume x lies on the x-axis in $\mathbb{R}^2$ as a simple case. Let $r = |x|$. Then the distance from the center of B_x to the point $(0,1)$ is given by

$$\sqrt{\left(\frac{r^2-1}{2r}\right)^2 + 1} = \sqrt{\frac{r^4 - 2r^2 + 1 + 4r^2}{4r^2}} = \frac{r^2+1}{2r}.$$

This proves the claim in dimension two, as the rest of the boundary of B_x lies to one side of the y-axis as well. This holds for arbitrary x by radial symmetry. In higher dimensions, the same argument holds to find the intersections between the boundaries, and about the rest of the intersection as well.

We will prove that there is no fixed $C > 0$ such that

$$\|Mf\|_{L^p} \leq C \|f\|_{L^p}$$

for all n and $f \in L^p$.

Pointiwse, we know

$$Mf_0(x) \geq \frac{1}{\mu(B_x)} \int_{B_x} |f_0| d\mu \geq \frac{1}{\mu(B_x)} \frac{v_n}{2}$$

by the above argument. Then

$$\|Mf_0\|_{L^p}^p \geq \int_{\mathbb{R}^n} \frac{1}{(\mu(B_x))^p} \frac{(v_n)^p}{2^p} = \int_{\mathbb{R}^n} \left(\frac{r^1+1}{2r}\right)^{np} d\mu = v_n \int_0^\infty \left(\frac{r^2+1}{2r}\right)^{np} r^{n-1} dr$$

after switching to polar coordinates. Note that $\|f_0\|_{L^p}^p = v_n$. Then all that is left is to show this integral diverges in n. This is true upon inspection with Desmos, but I could not show it rigorously.